

1

A: $64 * 13$

n	m	steps
64	13	64 is even
32	26	32 is even
16	52	18 is even
8	104	8 is even
4	208	4 is even
2	416	2 is even
1	832	1 is odd
		832 $64 * 13 = 832$

B: $60 * 13$

n	m	steps
60	13	60 is even
30	26	30 is even
15	52	15 is odd (+52)
7	104	7 is odd (+104)
3	208	3 is odd (+208)
1	416	1 is odd
		$416 + 52 + 104 + 208 = 780$ $60 * 13 = 780$

C: $59 * 13$

n	m	steps
59	13	59 is odd (+13)
29	26	29 is odd (+26)
14	52	14 is even
7	104	7 is odd (+104)
3	208	3 is odd (+208)
1	416	1 is odd
		$416 + 13 + 26 + 104 + 208 = 767$ $59 * 13 = 767$

#2

A = [100, 33, 22, 213, 65, 29, 153, 199, 47, 181, 85]

i = 0; j = 0

index = A[0]; pivot = 85

100 > 85: leave array alone

Greater than: [100]

Less than: []

i = 0; j = 0 += 1

A = [100, 33, 22, 213, 65, 29, 153, 199, 47, 181, 85]

i = 0; j = 1

index = A[1]; pivot = 85

33 < 85: **swap A[0] and A[1]**

A = [33, 100, 22, 213, 65, 29, 153, 199, 47, 181, 85]

i = 1

Greater than: [100]

Less than: [33]

i = 1; j = 1 += 1

A = [33, 100, 22, 213, 65, 29, 153, 199, 47, 181, 85]

i = 1; j = 2

index = A[2]; pivot = 85

22 < 85: **swap A[1] and A[2]**

A = [33, 22, 100, 213, 65, 29, 153, 199, 47, 181, 85]

i = 2

Greater than: [100]

Less than: [33, 22]

i = 2; j = 2 += 1

A = [33, 22, 100, 213, 65, 29, 153, 199, 47, 181, 85]

i = 2; j = 3

index = A[3]; pivot = 85

213 > 85: leave array alone

Greater than: [100, 213]

Less than: [33, 22]

i = 2; j = 3 += 1

A = [33, 22, 100, 213, 65, 29, 153, 199, 47, 181, 85]

i = 2; j = 4

index = A[4]; pivot = 85

65 < 85: **swap A[2] and A[4]**

A = [33, 22, 65, 213, 100, 29, 153, 199, 47, 181, 85]

i = 3

Greater than: [100, 213]

Less than: [33, 22, 65]

i = 3; j = 4 += 1

A = [33, 22, 65, 213, 100, 29, 153, 199, 47, 181, 85]

i = 3; j = 5

index = A[5]; pivot = 85

29 < 85: swap A[3] and A[5]

A = [33, 22, 65, 29, 100, 213, 153, 199, 47, 181, 85]

i = 4

Greater than: [100, 213]

Less than: [33, 22, 65, 29]

i = 4; j = 5 += 1

A = [33, 22, 65, 29, 100, 213, 153, 199, 47, 181, 85]

i = 4; j = 6

index = A[6]; pivot = 85

153 > 85: leave array alone

Greater than: [100, 213, 153]

Less than: [33, 22, 65, 29]

i = 4; j = 6 += 1

A = [33, 22, 65, 29, 100, 213, 153, 199, 47, 181, 85]

i = 4; j = 7

index = A[7]; pivot = 85

199 > 85: leave array alone

Greater than: [100, 213, 153, 199]

Less than: [33, 22, 65, 29]

i = 4; j = 7 += 1

A = [33, 22, 65, 29, 100, 213, 153, 199, 47, 181, 85]

i = 4; j = 8

index = A[8]; pivot = 85

47 < 85: swap A[4] for A[8]

A = [33, 22, 65, 29, 47, 213, 153, 199, 100, 181, 85]

i = 5

Greater than: [100, 213, 153, 199]

Less than: [33, 22, 65, 29, 47]

i = 5; j = 8 += 1

A = [33, 22, 65, 29, 47, 213, 153, 199, 100, 181, 85]

i = 5; j = 9

index = A[j]; pivot = 85

181 > 85: leave array alone

Greater than: [100, 213, 153, 199, 181]

Less than: [33, 22, 65, 29, 47]

i = 5; j = 9 += 1

A = [33, 22, 65, 29, 47, 213, 153, 199, 100, 181, 85].

i = 5; j = 10

index = A[10]; pivot = 85

swap A[10] and the pivot

A = [33, 22, 65, 29, 47, 85, 153, 199, 100, 181, 213]

Final Array:

A = [33, 22, 65, 29, 47, 85, 153, 199, 100, 181, 213]